SUVRADIP GHOSH

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SUMMARY

MCA Candidate experienced in building AI-enabled software by combining full-stack development, backend engineering, machine learning, LLMs, computer vision and system optimization to develop applications that solve real-world problems.

SKILLS

Programming Languages: Python, Java, JavaScript, C
Frontend: HTML, CSS, React.js, Tailwind CSS
Backend: Node.js, Express.js, FastAPI
Databases: MongoDB Atlas, SQL, SQLite
Tools & Platforms: Git, Docker, Clerk, Convex, ImageKit
AI/ML: PyTorch, TensorFlow, Scikit-Learn, OpenCV, SHAP

PROJECTS

Insight-ATS: AI-Assisted Hiring and Project-Based Evaluation Platform — Live | GitHub

Tech Stack: React.js, Recharts, FastAPI, PyTorch, BERT, RoBERTa, DistilBERT, SHAP

- Built an AI-powered ATS platform, **introducing contextual Project Complexity Analysis** and **semantic Resume–JD matching** beyond keyword-based approaches, using **3 transformer models** and SHAP explainability.
- Fine-tuned and evaluated 3 models: **BERT-NER** entity extractor (used 1000 resumes, **0.75 F1**), **RoBERTa** resume–JD semantic matcher (used 8000 resume–JD pairs, **on 0.78 Macro F1**), and **DistilBERT** project-complexity evaluator (**0.93 Macro F1**) using 3×5 repeated K-Fold validation.
- **Optimized multi-resume inference latency by 51.1%** (6.62s to 3.24s) by designing a FastAPI pipeline with lazy loaded singleton orchestration and in-memory cached model instances serving **4-dimensional** ATS scores.
- **Reduced API calls by 40%** using shared-JD batch screening; deployed on Hugging Face Spaces with Clerk auth.

CodeNest: AI-Powered Cloud IDE and Review Platform — Live | GitHub

Tech Stack: React.js, Express.js, MongoDB Atlas, Docker, Groq Llama-3.3-70B

- Built a **microservice-based** cloud IDE supporting **20+ concurrent users**, featuring isolated code execution in docker containers, AI code review, suggestions, flowchart visualization, code generation and **5-second** execution timeouts.
- Engineered a queue-based execution system with a **25-job buffer** and **2 worker processes** to prevent resource exhaustion and ensure reliable code execution under concurrent workloads.
- Integrated **Groq-hosted Llama-3.3-70B** for AI features, Mermaid flowchart generation, code analysis, and **Redis caching and API rate limiting**, achieving **1–3 second** response latency.
- Implemented **SHA-256 response caching**, a **20 requests/minute** rate limit, and a Redis-backed rate limiter with in-memory failover; deployed the platform and execution runner on Render.

Skin Disease Detection System Using CNN — GitHub

Tech Stack: Python, TensorFlow, EfficientNetB3, OpenCV, Scikit-Learn, CNN

- Developed **a skin disease recognition and classification model** on the HAM10000 dataset (**10,015** dermatoscopic images), restructuring labels into **3 classes**: benign, melanoma, and non-melanoma.
- Prevented data leakage through patient-id based train-validation-test splitting and used a dual-stream preprocessing pipeline using **CLAHE** enhancement and **black-hat morphology**-based hair removal.
- Fine-tuned **EfficientNetB3** using transfer learning, Focal Loss, Adam optimization, and **Test-Time Augmentation**.
- Achieved **90%** test accuracy with F1-scores of **0.89** (benign), **0.86** (melanoma), and **0.94** (non-melanoma), and evaluated performance using confusion matrices and per-class precision/recall analysis.

EDUCATION

Master of Computer Applications — 2024 – Present
Sikkim Manipal Institute of Technology — CGPA: **8.25**

Bachelor of Computer Applications — 2021 – 2024
Techno India University — CGPA: **8.39**

Higher Secondary (Class XII) — 2021
Maynaguri High School — Percentage: **88.6%**

CERTIFICATIONS

Android Developer Virtual Internship (Google & AICTE EduSkills) — Apr 2024 – Jun 2024
Grade: **Excellent** — Credential

IBM Professional Certificate: Full Stack Software Developer — Credential

IBM Professional Certificate: Data Science and AI — Credential